



CYBER SECURITY

(BCC-301)

Unit - 1 : INTRODUCTION TO CYBER CRIME

Complete Notes

Syllabus

Unit 1 : INTRODUCTION TO CYBER CRIME

Cybercrime- Definition and Origins of the word Cybercrime and Information Security,
Who are Cybercriminals? Classifications of Cybercrimes, A Global Perspective on
Cybercrimes, Cybercrime Era: Survival Mantra for the Netizens.

Cyber offenses: How Criminals Plan the Attacks, Social Engineering, Cyber stalking,
Cybercafe and Cybercrimes, Botnets: The Fuel for Cybercrime, Attack Vector

Cybercrime- Definition and Origin

Introduction -

Cybercrime refers to any illegal activity carried out using computers, digital devices, or computer networks. In such crimes, the computer may be:

- Target (e.g., hacking, data breach)
- Tool (e.g., online fraud, phishing)
- Medium/place (e.g., illegal online transactions)

It includes activities like:

- Hacking
- Identity theft
- Cyber stalking
- Online banking fraud

Origin -

The term “Cyber” is derived from the Greek word “Kybernetes”, meaning steersman or governor.

It was popularized by:

- Norbert Wiener
- In his book Cybernetics (1948)

Cybernetics means the study of control and communication in machines and living beings. Over time, the word “cyber” became associated with computers and the internet.

Historical Origin of Cybercrime -

(A) Early Stage (1960s–1970s) -

- Computers were large mainframes used by governments and research institutions.
- Early cybercrimes involved unauthorized access and data manipulation.
- Crimes were mainly committed by insiders.

(B) Internet Expansion (1980s–1990s) -

- Personal computers became common.
- Internet connectivity increased.
- The first major worm attack was the Morris Worm (1988).
- Hacking and software piracy increased.

(C) Modern Era (2000–Present) -

- Growth of e-commerce and online banking.
- Rise of phishing, ransomware, identity theft, and cyber terrorism.
- Cybercrime became global and organized.

Information Security_-

Information Security (InfoSec) refers to the protection of information and information systems from:

- Unauthorized access
- Disclosure
- Alteration
- Destruction

Objectives of Information Security_(CIA Triad)_-

1. Confidentiality – Information is accessible only to authorized persons.
2. Integrity – Information remains accurate and unaltered.
3. Availability – Information is available when required.

Information security helps in:

- Preventing cybercrime
- Protecting sensitive data

Who are Cybercriminals?

Cybercriminals are individuals or groups who use computers, digital devices, or networks to commit illegal activities in cyberspace for financial gain, revenge, political motives, or personal satisfaction.

Definition :

Cybercriminals are persons who intentionally use information technology to perform unlawful activities in cyberspace.

They may work individually, as insiders, or as part of organized international networks.

Types of Cybercriminals -

1. Hackers -

A hacker is a person who gains unauthorized access to computer systems.

Types :

- White Hat Hackers – Ethical hackers who test security systems.
- Black Hat Hackers – Malicious hackers who steal or destroy data.
- Grey Hat Hackers – Break rules but may not have criminal intent.

2. Crackers -

- Break into systems to steal, modify, or destroy data.
- Often involved in software piracy and password cracking.

3. Script Kiddies -

- Break into systems to steal, modify, or destroy data.
- Often involved in software piracy and password cracking.

4. Cyber Terrorists -

- Use cyber attacks to create fear or disrupt government systems.
- Target critical infrastructure like banking, defense, and power systems.

5. Organized Cybercrime Groups -

- Professional criminal networks operating internationally.
- Engage in large-scale fraud, ransomware attacks, and data theft.

6. Insiders -

- Employees or former employees who misuse authorized access.
- Often motivated by revenge or financial gain.

7. Phishers and Online Fraudsters -

- Trick users into revealing personal information.
- Conduct email scams and fake websites.

Classification of Cybercrimes -

Cybercrimes can be classified into the following categories:

1. Crimes Against Individuals -

These crimes target a specific person.

Examples:

- Identity theft
- Cyber stalking
- Online harassment
- Email spoofing
- Credit card fraud

2. Crimes Against Property -

These crimes damage or steal digital property.

Examples:

- Data theft
- Intellectual property theft
- Virus and malware attacks
- Ransomware

3. Crimes Against Organizations -

These target companies or institutions.

Examples:

- Data breaches
- Corporate espionage
- Website defacement
- Denial-of-Service (DoS) attacks



4. Crimes Against Government -

These threaten national security.

Examples:

- Cyber terrorism
- Hacking government websites
- Attacks on defense systems
- Cyber warfare



Global Perspective on Cybercrimes

Cybercrime is a borderless and transnational crime that affects individuals, organizations, and governments worldwide. With the rapid growth of the internet, digital banking, cloud computing, and e-commerce, cyber threats have become a global concern.

A global perspective on cybercrime means understanding its international nature, impact, and cooperative efforts to control it.

Nature of Cybercrime as a Global Threat -

Key Features:

- Borderless – Attackers can operate from any country.
- Anonymity – Criminals hide identity using VPNs, botnets, dark web.
- High Speed – Attacks spread worldwide in seconds.
- Organized Networks – International cybercrime syndicates operate across continents.

Global Impact of Cybercrime -

Cybercrime affects:

(A) Individuals :

- Identity theft
- Online fraud
- Social media hacking

(B) Businesses :

- Data breaches
- Financial losses
- Reputation damage

(C) Governments

- Cyber espionage
- Attacks on critical infrastructure
- Cyber warfare



Global Efforts to Control Cybercrime -

(A) International Organizations -

- INTERPOL – Coordinates international cybercrime investigations.
- United Nations – Works on global cyber policies.

(B) International Agreements -

- Budapest Convention on Cybercrime – First international treaty addressing cybercrime.

(C) National Cyber Laws -

Countries have enacted cyber laws, such as:

- Information Technology Act 2000 (India)

Cybercrime Era – Survival Mantra for Netizens

In the digital era, every internet user (netizen) must follow cybersecurity best practices to stay safe.

1. Strong Password Management :

- Use complex passwords.
- Enable two-factor authentication (2FA).
- Avoid sharing OTPs.

2. Safe Browsing Practices :

- Avoid clicking unknown links.
- Verify website URLs (https).
- Do not download from untrusted sources.

3. Protect Personal Information :

- Limit sharing on social media.
- Never disclose bank details or passwords.

4. Use Updated Security Tools :

- Install antivirus software.
- Regularly update operating systems and applications.

5. Be Alert Against Phishing :

- Do not trust suspicious emails or SMS.
- Check sender details carefully.

Cyber offenses: How Criminals Plan the Attacks

Cyber offenses are illegal activities committed using computers, networks, or digital devices. Cybercriminals do not attack randomly; they follow a systematic and planned approach to exploit vulnerabilities and achieve their objectives.

Cyber attacks are usually executed in stages, similar to a military operation.

Stages of Planning a Cyber Attack -

Cybercriminals generally follow a structured methodology:

1. Reconnaissance (Information Gathering) -

This is the first stage where attackers collect information about the target.

Methods include:

- Social media profiling
- Company website analysis

- Open-source intelligence (OSINT)
- Email harvesting

Objective: Identify vulnerabilities and weak points.

2. Scanning and Enumeration -

- Scanning networks to find open ports.
- Identifying active systems and services.
- Detecting software versions and security gaps.

Attackers use automated tools to locate vulnerabilities.

3. Gaining Access (Exploitation) -

- Using malware, phishing, or password attacks.
- Exploiting software vulnerabilities.
- Bypassing authentication systems.

Common methods:

- Brute-force attacks
- SQL injection
- Ransomware deployment

4. Maintaining Access -

After entering the system, attackers try to:

- Install backdoors
- Create hidden user accounts
- Deploy Trojans
- Connect to command-and-control (C2) servers

Purpose: Maintain long-term control over the system.

5. Covering Tracks -

- Deleting log files
- Encrypting stolen data
- Using proxy servers or VPNs
- Hiding identity via dark web

Objective: Avoid detection and legal action.

6. Motives Behind the Attack -

Cybercriminals plan attacks for:

- Financial gain
- Data theft
- Corporate espionage
- Political or ideological reasons

Social Engineering

Social Engineering is a technique used by cybercriminals to manipulate people into revealing confidential information or performing actions that compromise security. Instead of attacking technical systems directly, attackers exploit human psychology. Social Engineering is the art of manipulating individuals to gain unauthorized access to information or systems.

Why Social Engineering is Effective -

- Humans are the weakest link in cybersecurity.
- People tend to trust authority or urgent requests.
- Lack of awareness makes users vulnerable.
- Emotional triggers (fear, greed, curiosity) are exploited.

Common Techniques of Social Engineering-

1. Phishing.-

- Fraudulent emails or messages pretending to be from trusted sources.
- Objective: Steal passwords, OTPs, bank details.
- Example: Fake bank email asking to update account details.

2. Pretexting.-

- Creating a fake identity or scenario to obtain information.
- Example: Attacker pretending to be a bank officer or IT support staff.

3. Baiting.-

- Offering something attractive (free software, USB drive).
- Victim unknowingly installs malware.

4. Tailgating (Piggybacking).-

- Unauthorized person enters restricted area by following an authorized person.

5. Vishing and Smishing.-

- Vishing – Voice call scams.
- Smishing – SMS-based phishing attacks.

Stages of a Social Engineering Attack -

1. Information Gathering – Collecting victim details from social media.
2. Building Trust – Creating believable communication.
3. Exploitation – Extracting confidential information.
4. Execution – Using obtained data for fraud or hacking.

Prevention of Social Engineering -

- Do not share passwords or OTPs.
- Verify sender identity before responding.
- Use two-factor authentication (2FA).

Cyber Stalking

Cyber stalking is a form of online harassment in which a person uses the internet, social media, emails, or other digital platforms to threaten, harass, or repeatedly monitor another person.

Cyber stalking is the act of using electronic communication to repeatedly harass or threaten an individual, causing fear or emotional distress.

It is a serious cyber offense and punishable under cyber laws.

Features of Cyber Stalking:-

- Repeated and unwanted communication
- Monitoring victim's online activities
- Sending threatening or abusive messages
- Creating fake profiles
- Posting false or defamatory content

Common Methods Used in Cyber Stalking -

- Social Media Harassment – Continuous abusive messages or comments.
- Email Threats – Sending threatening or obscene emails.
- Fake Profiles – Creating fake accounts to defame or trap victims.
- Monitoring & Tracking – Tracking activities using spyware or GPS.
- Doxxing – Publishing personal information online without consent.

Motives Behind Cyber Stalking -

- Personal revenge
- Obsession or jealousy
- Rejected relationships
- Harassment for fun
- Psychological disturbance

Cybercafe and Cybercrimes

A cybercafe is a public place that provides internet access to users for browsing, email, gaming, online transactions, etc. Since multiple unknown users access the same systems, cybercafes can become hotspots for cybercrimes if not properly regulated. Cybercafes may be misused by criminals to perform illegal online activities anonymously.

Why Cybercafes are Vulnerable to Cybercrime -

- Public access computers
- Anonymous users
- Shared IP addresses
- Weak monitoring systems
- Lack of proper identity verification

These factors make tracing criminals difficult.

Types of Cybercrimes Committed through Cybercafes -

1. Hacking and Unauthorized Access -

Criminals use cybercafe computers to hack accounts or networks.

2. Online Fraud and Phishing -

Sending fraudulent emails or creating fake websites.

3. Identity Theft -

Accessing others' personal accounts using public systems.

4. Cyber Terrorism -

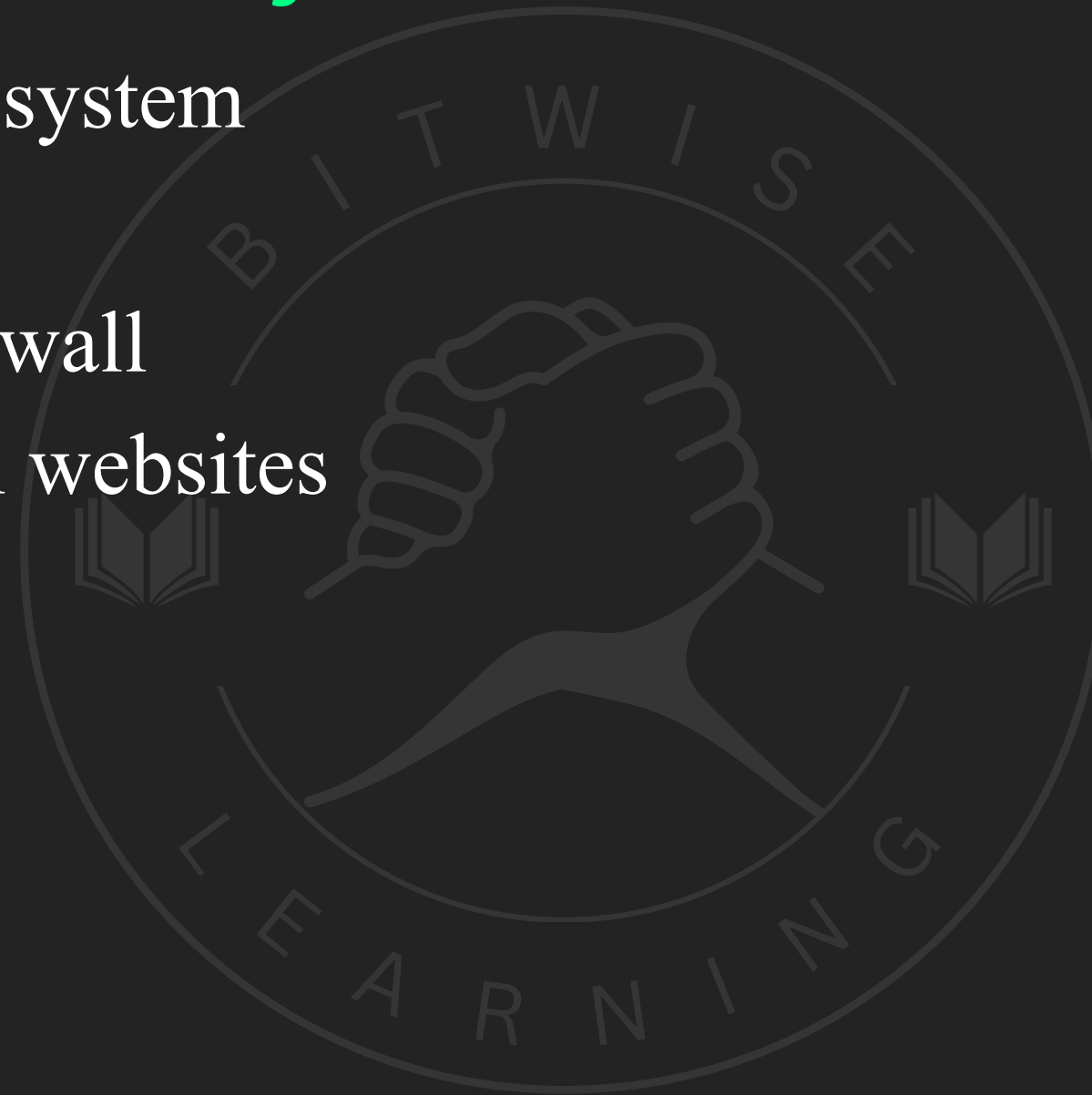
Sending threatening emails or planning illegal activities.

5. Malware Distribution -

Downloading and spreading viruses through public systems.

Preventive Measures for Cybercafe Owners -

- Proper user registration system
- Maintain log records
- Install antivirus and firewall
- Restrict access to illegal websites
- Regular system updates



Botnets: The Fuel for Cybercrime

A Botnet is a network of infected computers or devices (called bots or zombies) that are remotely controlled by a cybercriminal without the owner's knowledge.

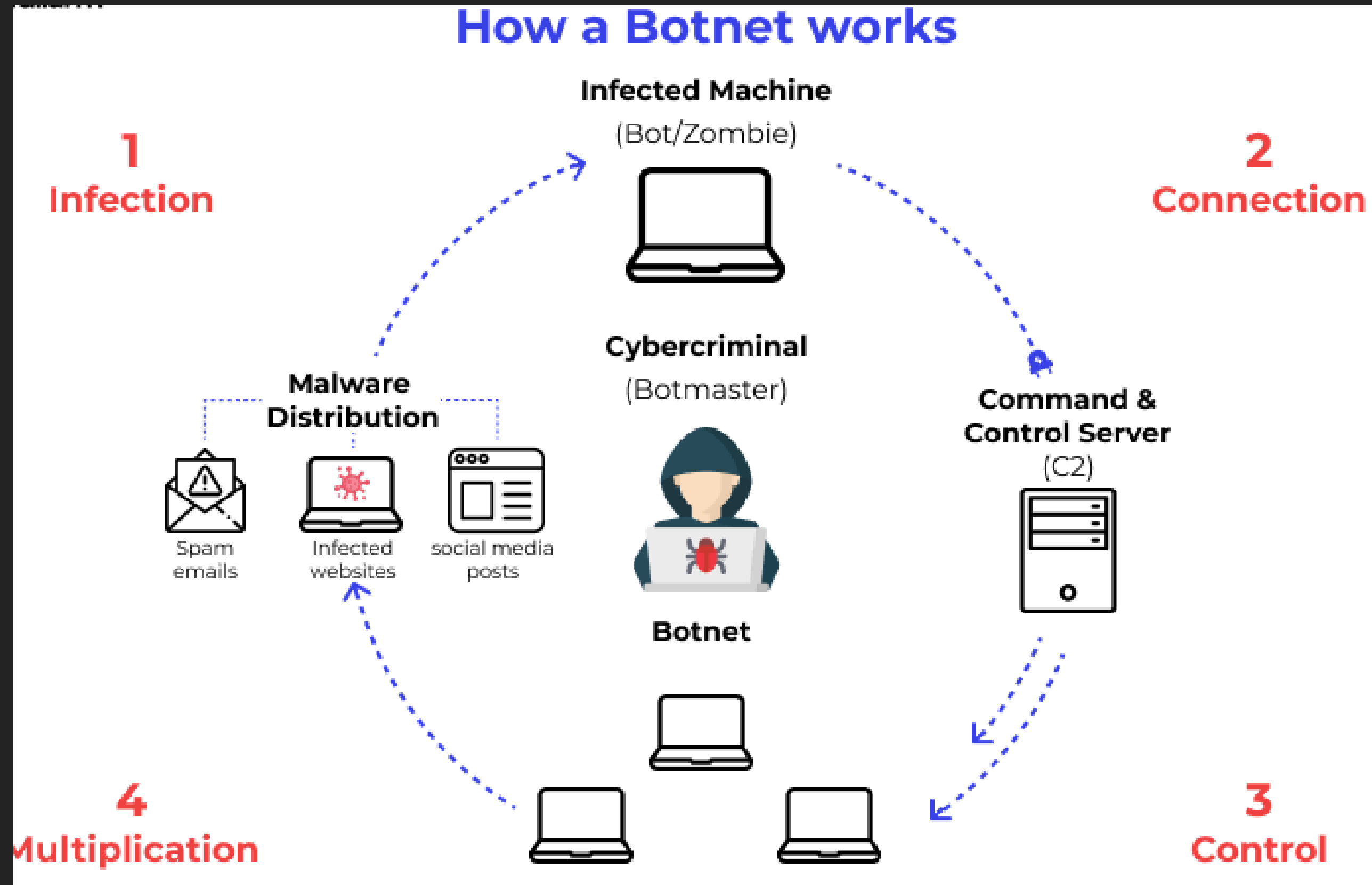
Botnets act as a powerful tool that enables large-scale cyber attacks, making them the “fuel” for cybercrime.

The attacker who controls the botnet is known as a Botmaster.

Devices in a botnet may include:

- Computers
- Smartphones
- IoT devices (CCTV cameras, routers, smart devices)

Structure of a Botnet -



Structure of a Botnet - Components:

1. Botmaster (Attacker) – Controls the network.
2. Command and Control (C&C) Server – Sends instructions to bots.
3. Bots/Zombies – Infected devices executing commands.

Working:

- Malware infects devices.
- Devices connect to the C&C server.
- Botmaster sends commands to perform attacks.

How Botnets are Created -

1. Spreading malware via phishing emails.
2. Exploiting software vulnerabilities.

3. Malicious downloads or fake links.
4. Infecting unsecured IoT devices.

Uses of Botnets in Cybercrime -

Botnets are widely used for:

1. Distributed Denial of Service (DDoS) Attacks -

Overloading a website with traffic to shut it down.

2. Spamming -

Sending bulk spam emails.

3. Data Theft -

Stealing login credentials and personal information.

4. Click Fraud -

Generating fake ad clicks for financial gain.

5. Cryptocurrency Mining -

Using infected systems to mine digital currency.

Why Botnets are Dangerous -

- Difficult to detect & Operate globally
- Large-scale coordinated attacks
- Exploit thousands or millions of devices
- Cause huge financial and reputational damage

Prevention Measures -

- Install updated antivirus software.
- Keep operating systems updated.
- Avoid suspicious links and downloads.
- Secure IoT devices with strong passwords.
- Use firewalls and intrusion detection systems.

Attack Vector

An Attack Vector is the path or method used by a cybercriminal to gain unauthorized access to a computer system, network, or data.

An attack vector is the technique or route through which a cyber attack is carried out.

Attack vectors exploit weaknesses in:

- Software
- Hardware
- Networks
- Human behavior

Purpose of an Attack Vector -

The main objectives are:

- Unauthorized access
- Data theft

- Malware installation
- System disruption
- Financial fraud

Common Types of Attack Vectors -

1. Phishing Attacks:

- Fraudulent emails or messages.
- Trick users into revealing passwords, OTPs, or bank details.

2. Malware:

- Viruses, worms, Trojans, ransomware.
- Installed through malicious downloads or email attachments.

3. Weak Passwords:

- Brute-force attacks.
- Password guessing or credential stuffing.

4. Unpatched Software:

- Exploiting software vulnerabilities.
- Using outdated systems without security updates.

5. Insider Threats:

- Employees misusing authorized access.
- Data leakage or sabotage.

6. Network-Based Attacks:

- Man-in-the-Middle (MITM) attacks
- Distributed Denial of Service (DDoS)

7. Physical Access:

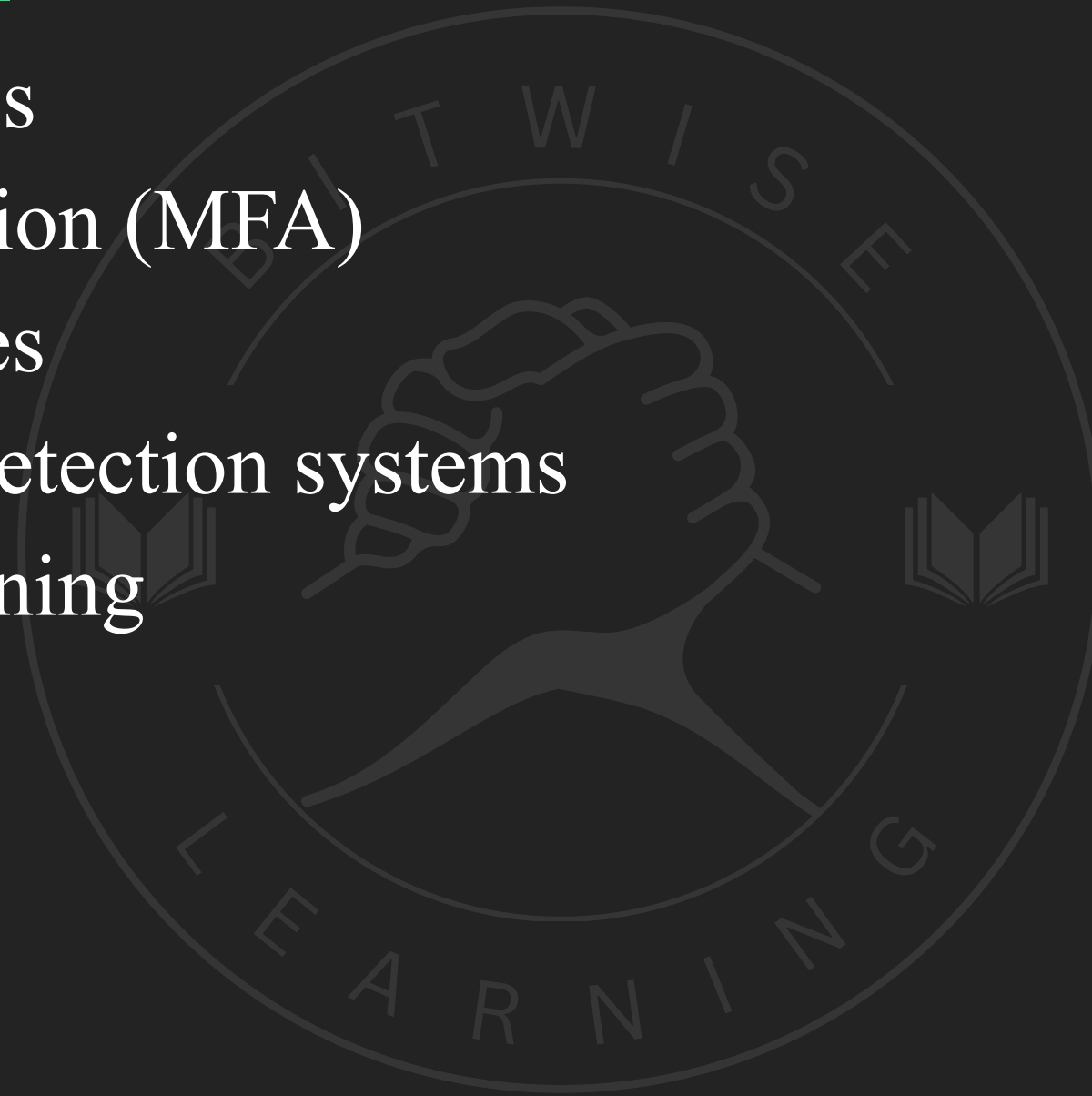
- Stolen devices
- Unauthorized USB drives
- Tailgating in secure areas

Why Attack Vectors are Dangerous-

- Easy exploitation of human errors
- Difficult to detect initially
- Can cause financial and reputational damage
- May lead to large-scale cyber attacks

Prevention Measures -

- Strong password policies
- Multi-factor authentication (MFA)
- Regular software updates
- Firewall and intrusion detection systems
- User awareness and training



THANK
you!